

PRITHVEX System Architecture

End-to-End Workflow & Dashboard Design · Team Envirex

1. End-to-End Workflow

PRITHVEX is organised as a five-stage end-to-end workflow, with three shared system layers running underneath every stage.

Stage	Components	Outcome
1. Data collection & integration	IoT sensors · ERP systems · third-party feeds	Sources onboarded
2. Uncertainty calibration & risk scoring	Bayesian inference · feature engineering · ECE evaluation	Model trained, +30% accuracy
3. Dashboard monitoring & alerting	React frontend · WebSocket streaming · calibrated thresholds	Live heatmaps, ~40% faster response
4. Scenario simulation & recommendations	Monte Carlo runs · what-if branching · auto risk-adaptation	Decision suggestions, ~25% lower holding cost
5. Decision implementation & feedback loop	Cloud integration · model recalibrated · continuous learning	Adaptive self-improving model

Underlying system layers: cloud infrastructure (AWS / Azure / GCP / Kubernetes), the Bayesian core (probabilistic, uncertainty-quantification engine), and the simulation engine (Monte Carlo scenario planning).

2. Dashboard Panel Architecture

Four live panels sit on top of the shared pipeline and system layers, each with its own internal data flow.

2.1 Risk Heatmap Panel

Live data feeds (IoT, ERP, third-party APIs) stream through a WebSocket connection for sub-second ingestion into the risk scoring engine, which assigns a Bayesian confidence score per node. A React-based heatmap renderer visualises geographic risk zones, while an alert threshold check runs in parallel; a calibrated threshold breach triggers a push notification to the risk team. The dashboard then refreshes continuously in a live update loop. Reported outcome: roughly 40% faster disruption response time.

2.2 Scenario Simulation Panel

A user triggers a scenario by selecting variables and risk parameters. The Monte Carlo engine then runs thousands of probabilistic scenarios and groups the outcomes into best-case, base-case, and worst-case branches. Reported outcome: automated, risk-aware recommendations generated per branch.

2.3 Uncertainty Report View

The Bayesian inference engine runs on live supply chain data to generate confidence-bounded forecasts per supplier, route, and warehouse node. An Expected Calibration Error (ECE) metric compares predicted confidence to actual outcomes: a low ECE marks the model as reliable, while a high ECE triggers recalibration on new data. Both paths converge on a reliability diagram rendered as a fully auditable, explainable output. Reported outcome: explainable calibrated intelligence with no black-box scoring.

2.4 Real-Time Alert Panel

The risk scoring engine runs continuously across all monitored nodes. A threshold breach fires a calibrated trigger condition, which feeds an alert generation engine that scores severity and flags affected nodes. A push notification is dispatched to the risk team instantly; once the user acknowledges and acts, the action is logged and the outcome is fed back into the model, closing the loop. Reported outcome: proactive risk mitigation with roughly 40% faster response time.

3. Solution Architecture (Platform View)

At the platform level, ENVIREX wraps PRITHVEX in six functional blocks that map directly onto the workflow above:

Block	Technology
Data Ingestion	ERP / APIs / Satellites / IoT
ML Ensemble	XGBoost · LightGBM · LSTM · Transformers
Conformal Prediction	MAPIE-based uncertainty intervals
SHAP Attribution	Evidence trails for every risk score
Network Intelligence	Tier-1/2/3 Monte Carlo propagation
Action Playbook	AI-ranked mitigation strategies

The platform is designed to complete every step, from ingestion to recommendation delivery, in under three seconds, with zero cloud dependency and full reproducibility.

4. Technology Stack Summary

- AI / ML core: XGBoost, LightGBM, scikit-learn, pandas, NumPy, LSTM, Transformers.
- Uncertainty engine: MAPIE conformal prediction, KL-divergence drift monitor, adaptive recalibration.
- Explainability: SHAP feature attribution with plain-language risk alerts.
- Dashboard: Streamlit, Plotly, NetworkX, Monte Carlo simulation engine; production panels use a React frontend with WebSocket streaming.
- Infrastructure: zero cloud dependency, fully reproducible, CSV and API data ingest, with optional AWS/Azure/GCP/Kubernetes deployment.

5. Deployment Workflow

Step	Description
1	Data ingestion from ERP, logistics APIs, IoT sensors, and supplier systems
2	PRITHVEX forecasting engine predicts inventory pressure across all SKUs
3	Risk simulation evaluates disruption severity with Monte Carlo propagation
4	PRITHVEX classifies operational risk with conformal confidence intervals
5	Recommendation engine generates ranked mitigation actions with ROI quantification

6

Command Center dashboard visualises all analytics, alerts, and decision support